Summary

Growth occurs by cell division, cell elongation and cell maturation. The first phase is lag phase, the second is log phase and the final phase is steady state phase. The log phase is otherwise known as **exponential phase**. The three phases are collectively called Grand period of growth. Plant growth and development are controlled by both internal and external factors. The internal factors are chemical substances called Plant Growth Regulators (PGRs). The hormones are classified into five groups: Auxins, gibberellins, cytokinins, abscisic acid and ethylene. These PGRs are synthesized in various parts of the plant. PGRs may act synergistically or antagonistically. Mechanism of flowering is controlled by light period (photoperiodism) and temperature (vernalization). The physiological changes on flowering with effect from relative length of light and darkness (photoperiodism) are called photoperiodism. A bluish biliprotein responsible for the perception of light in photophysiological process (induction and inhibition of flowering) is called **Phytochrome**. Besides photoperiod certain plants require a low temperature in the earlier stages for flowering. Many biennial and perennial plants are induced to flower by low temperature (0°C to 5°C). This process is called **vernalization** and the reversal effect of vernalization is called **devernalization**. The condition of a seed when it fails to germinate even in suitable environmental condition is called **seed dormancy**. Thus, dormancy can be overcome by following methods such as scarification, impaction, stratification, alternating temperatures and light. Senescence refers to all collective, progressive and deteriorative processes which ultimately lead to complete loss of organization and function. Senescence is of four types and they are overall, top, deciduous and progressive. Senescence is controlled by plant's own genetic programme. Death of the plant or its parts consequent to senescence is called **Programmed Cell Death** (PCD). The final stage of senescence is abscission. Abscission is a physiological process of shedding of organs from the parent plant body.

Evaluation

1. Select the wrong statement from the following:
   a. Formative phase of the cells retain the capability of cell division.
   b. In elongation phase development of central vacuole takes place.
   c. In maturation phase thickening and differentiation takes place.
   d. In maturation phase, the cells grow further.

2. If the diameter of the pulley is 6 inches, length of pointer is 10 inches and distance travelled by pointer is 5 inches. Calculate the actual growth in length of plant.
   a. 1.5inches b. 6 inches
   c. 12 inches d. 30 inches

3. _____ is the powerful growth inhibitor
   a. Ethanol b. Cytokinins
   c. ABA d. Auxin

4. Select the correctly matched one

A) Human urine	i) Auxin –B
B) Corn gram oil	ii) GA_3
C) Fungus	iii) Abscisic acid II
D) Herring fish sperm	iv) Kinitin
E) Unripe maize grains	v) Auxin A
F) Young cotton bolls	vi) Zeatin

   a) A-iii, B-iv, C-v, D-vi, E-i, F-ii,
   b) A-v, B-i, C-ii, D-iv, E-vi, F-iii,
   c) A-iii, B-v, C-vi, D-i, E-ii, F-iv,
   d) A-ii, B-iii, C-v, D-vi, E-iv, F-i

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